

The 5-Cs model — by role

Category	CS student's core purpose is learning	Math student's core purpose is learning.	Teacher's core purpose is teaching
Core (header note)	Relevant skills: 15 CS skills + content knowledge, e.g. programming language	Relevant skills: Process goals Content goals	providing a safe learning environment; providing meaningful inputs and/or resources; crafting appropriately-scaffolded assignments; assessing student progress and understanding; providing timely, meaningful feedback; targeting support to student skills and gaps
R · Reflect <i>Am I reflecting on what I learned, did wrong, did right, or had skill gaps in? Can I state my learning before engaging GA to sharpen, clarify, or restate that learning? Is the reflective insight mine?</i>	A CS student is reflecting when they are working on guidesheets, mindset reflections, thinking about their collaboration habits and use of class time, reflecting on common errors	A Math student is reflecting when they are working on refining their study technique, thinking about their collaboration habits and use of class time, reflecting on common errors from a test.	A teacher is reflecting when they are working on refining their pedagogy and teaching practice, creating professional learning/growth goals, reading through student surveys
C1 · Contribution: Create first, Sift in Dialogue <i>Am I outsourcing what my contribution is? Am I still visible in the final output?</i>	A CS student's contribution is their unique concept for a program, system, algorithm.	A Math student's contribution is their way of approaching a problem, the way they first make contact or approach an elegant proof.	A teacher's contribution is their way of approaching a course, topic, or lesson— the way they ground their content knowledge in their own experience and relationship to their students.
C2 · Core Purpose: Attempt – Ask - Audit <i>After asking for GA, did I audit the results? Did I ask for revisions, reject suggestions, or clarify my understanding?</i>	A CS student learns when they attempt to program, write, refactor, test, or design before consulting GA and consult only after getting stuck and running out of ideas.	A Math student learns when they try to solve the problem before consulting GA, and consult only after getting stuck and running out of ideas.	A teacher teaches when they relate to students, foster community, and envision the materials, assignments, assessment, and support before asking GA.
C3 · Clarification: Zoom to scope <i>After asking for GA, has the solution been found, or do I engage my own understanding with GA's scoping?</i>	A CS student seeks clarification when they attempt to debug, find errors in programs, diagrams, writing, and designs using exemplars, reasoning, and tools to locate issues. GA can be used to locate those attempts in the right search space.	A Math student seeks clarification when they attempt to check their work, determine why their answer is incorrect, they got stuck, etc. Math students use references, exemplars, tools like Desmos and calculators, and study guides to locate issues. GA can be used to locate those attempts in the right search space.	A teacher seeks clarification when they attempt to understand why a student or group of students did not understand the lesson, assignment, or resource provided. They use experience, colleagues, models of student learning, and relevant class data. GA can be used to locate those attempts in the right search space.
C4 · Correction: Redirect to Core Purpose <i>After finding the problem, did I use my own skills to address the correction or revision?</i>	A CS student corrects themselves and their work when they use the CS skills and content knowledge to fix the error found during clarification.	A Math student corrects themselves and their work when they use their process goals and content skills to fix the error found during clarification.	A teacher corrects themselves when they reteach, reassess, or provide additional materials using their core skills to tackle the error found during clarification
C5 · Cost Lighten the Load <i>Is this reducing load outside my core purpose and/or increasing my capacity to handle something *inside* the core purpose?</i>	A CS student reduces cost when they reduce the cognitive load or complexity of project planning and understanding non-critical parts of the project (e.g. artistic design)	A Math student reduces cost when they reduce the cognitive load of the problem, outside the core purpose by planning to study for a test, generating problems to attempt, or understanding unknown vocabulary.	A teacher reduces cost when they reduce the administrative load of teaching: planning, uploading, downloading, data collation and entry, and mass communication.